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Last Words

Justify Your Prompts!

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When you use a large language model (LLM) in your research, you often need to formulate a prompt to elicit some relevant output from the LLM. This step is challenging since (1) LLMs are known to be brittle and their results may vary drastically between different prompts; and (2) for any given task, there are infinitely many possible prompts. Thus we end up with the following problem: If you cannot try out infinitely many prompts, how do you justify your selected prompt? This article discusses different ways in which authors may motivate their choices, and urges authors to consider how their prompting strategy might be justified.

1. Introduction

The introduction of ChatGPT (OpenAI 2022) has clearly been a watershed moment. In the Before Times, whenever we wanted a language model to be able to perform a particular task, we would take a pretrained model and fine-tune it on a dataset exem-

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plifying the task-to-be-performed. We knew that the generalizability of the model and our findings depended on the training data and the benchmark. Nowadays, however, a very large, pretrained, language model may be made to carry out the desired task through **prompting**: specifying the task to be carried out through natural language.[1,2]

But what does this mean for the generalizability of our results? As the prompting search space is (effectively) infinite, any large language model (LLM) related research using prompts necessarily involves sampling: No study is able to evaluate the whole search space of prompts. This is not a new problem, however: Almost any research involves numerous *researcher degrees of freedom* which, if inadequately controlled for and documented, can lead researchers to false conclusions (Simmons, Nelson, and Simonsohn 2011; Wicherts et al. 2016). While much recent work focused on evaluating automatically generated texts has emphasized the importance of choosing the right automated metrics (Schmidtová et al. 2024) and human evaluation methods and providing adequate documentation to improve reproducibility (Belz et al. 2023), these are far from being the only degrees of freedom. Research using LLMs with prompting generally does not follow similar practices, despite the fact that an LLM's performance on a given task varies greatly with the chosen prompt. Subtle differences in prompts can dramatically impact LLM performance: changing prompt formatting (He et al. 2024), spelling (Schmidtová et al. 2025), phrasing (Lunardi et al. 2025), information structure. (Brucks and Toubia 2025), etc., all impact task performance. Beyond these issues, language itself is infinitely productive,[3] with endless ways to ask a model to perform a particular task. This brittleness of performance and variability of valid prompts necessitate proper documentation of the prompt sampling and selection process in order for others to evaluate the reliability (and potential generalizability) of the study.

Discussions of how to decide and report on sampling strategies are plentiful across scientific fields.[4] Perhaps most relevant for studies involving the creation of new software systems, the SIGSOFT reporting standard for software engineering research (Ralph et al. 2021) states that it is essential for any empirical study involving sampling to explain (a) the goal of the sampling, (b) the sampling strategy, (c) why the selected strategy is reasonable for the selected goal, and (d) the reasoning behind the selection of study objects. In this article, we (1) contextualize prompt justification as part of the larger metascientific discourse on reproducibility, (2) discuss different kinds of justifications that researchers might use and when these justifications may be appropriate, and (3) argue for the necessity of strong justifications and adequate documentation to ensure scientific rigor, improve reproducibility, and enable the collaborative development of generalizable findings. Our article contributes to a long history of metascientific discussion in the computational linguistics and natural language processing community. Earlier Last Words segments covered whether and how to compete with big tech (Kilgarriff 2007), how to benefit and build upon research in other fields

1 The term "prompting" is overloaded; it has different meanings depending on the context. In this article, we consider "prompts" to be the initial instructions that tell the system what to do. For ease of presentation, we ignore the more complex case where systems consist of a model and a prompt, and users interact with that system through further prompts.

2 Although some research also considers embeddings as *continuous prompts*; see Liu et al. (2023) for discussion.

3 This relates as well to general problems with evaluating automatically generated texts: For all but the simplest cases there are many, many linguistically valid ways to complete a task (i.e., express the same meaning), which makes it impossible for a set of references to cover the full range of "correct" answers.

4 For example, in economics (Salle and Yıldızoğlu 2014), journalism (Firdaus, Aksar, and Gong 2024), medicine (Msaouel 2023), and psychology (Shen et al. 2011).

(Reiter 2007; Spärck Jones 2007; Wintner 2009; Krahmer 2010, particularly linguistics and psychology), reproducibility (Pedersen 2008), applications and impact (Belz 2009), the importance of negative results (Szpakowicz 2010), data ethics and integrity (Fort, Adda, and Cohen 2011), terminological clarity and epistemic soundness (Riezler 2014), and best practices for developing shared tasks (Nissim et al. 2017).[5]

2. Background

Before we consider prompt justification strategies, we first provide pointers to the literature on prompting strategies and prompting quality, after which we discuss related work in metascience for computational linguistics and the sampling justification strategies that are used in the psychology literature. These strategies form the inspiration for the prompt justification strategies listed herein.

2.1 Prompting Strategies

Many prompting strategies have been developed in the literature. Schulhoff et al. (2025) provide a comprehensive taxonomy, where 58 different techniques have been identified, organized in six main categories. A concurrent survey by Sahoo et al. (2025) found 41 different prompting techniques, organized by application domain. This gives prompt authors some sense of when to use which set of prompting techniques, although the number of options remains daunting. An earlier survey by Liu et al. (2023) offers a more modest taxonomy of prompting methods, but it is just constructed at a higher level of abstraction (e.g., by having "hand-crafted prompt templates" as the lowest level, and not distinguishing different ways to formulate a prompt). Finally, the survey presented in Vatsal, Dubey, and Singh (2025) focuses on multilingual prompt engineering, covering 39 prompting techniques applied to 30 multilingual NLP tasks.

Of course, these different techniques can also be combined in different ways, leading to a combinatorial explosion. So it is natural to ask: With all these prompting techniques in mind, how can we write a good prompt?

2.2 Prompt Quality

We are starting to see more and more evidence-based guidelines on prompt writing. For example, Long et al. (2025) carried out a thorough study on the effectiveness of different prompting techniques for different NLP tasks, resulting in a set of recommendations for prompt authors. Other guidelines include, for example, those provided by Maaz et al. (2024) and Geroimenko (2025). But even with recommendations or prompt assessments at hand, there are still *many* different ways to formulate your prompt. Thus the question arises: How do you justify the exact formulation of your prompt?

5 Other related studies focus on, for example, best practices for studying usability (Dybkjær and Bernsen 2001) and developing language resources (Arranz et al. 2008) or how to adapt to changing methodologies (Saphra et al. 2024). Not to mention papers such as Church (2005), which do not address research practices but the practices that apply to publication and review.

3. Justification Strategies

As a general research principle, for any kind of study, researchers should always be able to motivate their design choices.[6] We take our inspiration from the psychology literature, where studies have been published on ways to justify sample sizes (Lakens 2022) and alpha values for statistical significance testing (Lakens et al. 2018). Taking the former as an example, Lakens (2022) identifies different strategies that a researcher might use to justify their choice of sample size. We may paraphrase these as follows:

1. *Avoid the problem altogether* by using the entire population.
2. *Appeal to resource constraints* as a reason for a smaller sample.
3. *Compute an optimal solution* by using a power analysis or other statistical tools to obtain a sample size that is large enough to draw valid conclusions, but not larger than needed.[7]
4. *Use heuristics* such as sample sizes mentioned in the literature, or "rules of thumb" mentioned by experienced researchers.
5. *No justification.* The researcher honestly reports not to have any justification.

These justification strategies have clear parallels with the justification strategies that may be used to motivate prompting decisions. Applying their categorization scheme to this domain results in the justification types discussed below. (In practice, these different types may often be combined.)

3.1 Avoiding the Problem or Embracing the Variation

Avoiding the problem in the sense described by Lakens (2022)—by using the whole population—is not feasible. It is literally impossible to test every possible prompt, and any selection of prompts will by necessity involve sampling. However, researchers might still choose to "avoid" the problem by simply acknowledging it and using multiple prompts, documenting the chosen prompts and the resulting variation. This approach is advocated by Mizrahi et al. (2024), who compared the performance of 20 different LLMs across 39 tasks. Instead of using a single set of instructions, the authors used a large set of instruction paraphrases. They show that both manual and automatic paraphrases lead to a high degree of variation in model performance. Based on their results, the authors argue that LLMs should be evaluated using a large collection of prompts and reporting the *average performance*, documenting the range of potential

6 For example, the Netherlands Code of Conduct for Research Integrity (applicable to all researchers in the country) specifically instructs researchers to "Ensure that the methods you employ are well justified" (KNAW et al. 2018). More generally, The European Code of Conduct for Research Integrity requires researchers to "design, carry out, analyze, and document research in a careful, transparent, and well-considered manner" (ALLEA 2023, p. 7).

7 We have merged two items from the original study (*Accuracy* and *Power analysis*) since the distinction is not relevant to us here.

outcomes.[8,9] Although this strategy is elegant in the way it embraces variation, it does run into three issues:

1. Because there are infinitely many possible prompts, you still need to make some kind of selection. Both the selection procedure (e.g., an initial prompt and the generation of paraphrases) and the sample size still need to be justified.
2. This approach is computationally expensive, since the number of computations linearly increases with the number of paraphrases.
3. Depending on the task, it may not be feasible to use multiple paraphrases, since the LLM outputs may be used for downstream processes, where the amount of effort (super)linearly increases with the number of outputs.

With these observations we do not wish to dispense with the multiple prompts approach; we only want to highlight the complexity of the situation. Critically analyzing this approach leads to interesting new questions, such as: How should one compose the set of prompts that is used for this approach,[10] and how large should that set be?

3.2 Resource Constraints and Reasonable Effort

Resources like time and money are finite and as a natural consequence researchers must make understandable compromises.[11] Although it is hard to draw the line, authors might appeal to the *reasonable effort heuristic*: As a community, we expect the authors to put in a reasonable amount of effort to develop the prompts for their study (but it is not clear how much that should be, exactly). Generally speaking, the first prompt that comes to mind may not be optimal, but it is also not desirable to spend years developing the optimal prompt. So, it is important to show your work, describing the iterative process through which the prompts came about. If the results of the study hinge on the quality of the prompt, it is important to mention this as a limitation of the study.

3.3 Evidence-based Justification

To demonstrate whether a particular prompt leads to plausibly good results for a given LLM there needs to be some form of evidential proof. We have identified three particular subtypes of justifications that authors could use: showing that the results are good enough; showing that the prompts are optimal; and arguing for prompts on the basis of representativeness.

8 This also avoids researchers cherry-picking the results that favor their narrative.
9 As a side note, this approach also shows parallels with *multiverse analyzes* that are carried out in psychological research (e.g., Steegen et al. 2016), Mazei et al. 2025). The idea here is that, instead of carrying out only a single analysis, the researcher explore multiple different analyzes based on different reasonable assumptions (e.g., using different exclusion criteria, or different ways to discretize a continuous variable).
10 For example, could we use one LLM to generate a prompt for another LLM in the same model family (or even having LLMs generate prompts for themselves, e.g., as in Nye, Mee, and Core 2023).
11 For a discussion of barriers in NLP research, see van Miltenburg et al. (2023).

3.3.1 Results Are Good Enough. If the results using a particular prompt are good enough for your purposes, then it may not be relevant whether the prompts could have been improved. Of course, this shifts the problem from justification to demonstration: How can you demonstrate that the outputs are good enough? One option to consider is to carry out a *pilot study* where a representative number of outputs is assessed according to some relevant quality metric (Sripada, Reiter, and Hawizy 2005; van Miltenburg et al. 2021). This strategy may not be appropriate when the goal of your study is to show how good LLM outputs can be. But it may be all you need if the LLM outputs only form part of the research process, where you build on the outputs toward a larger goal.

3.3.2 Prompts Are (Locally) Optimal. If there is an objective way to assess the quality of the output, then it may be feasible to compute an optimal prompt for a given LLM-dataset combination. Indeed, the survey by Liu et al. (2023) discusses two general strategies that LLM researchers may use: automatically discovering discrete (Natural Language) or continuous (embeddings) prompts.[12] For example, Zhou et al. (2023) show for 21 (out of 24) different tasks that LLMs achieve comparable or better performance in formulating prompts than human annotators. Moreover, Do et al. (2025) present an approach to automatically select the optimal prompt from a finite set of synthetic candidate prompts, based on clustering and a prompt evaluator that scores the prompts and selects the best one. In theory, given the infinite search space, it may still be possible to formulate a better prompt, but this is the closest we can get to an optimal prompt. There are two clear limitations to this justification approach:

1. As noted above, this approach may not be feasible for tasks where the quality of the output is hard to assess.
2. This justification approach is only accessible to a select group of people with the technical abilities to set up the software to find the optimal prompt for the task at hand. Future researchers may want to find ways to reduce this barrier.

Because most optimization strategies offer *locally* optimal solutions, researchers should be careful in their justification not to imply that these solutions are also *globally* optimal (i.e., the best possible prompts).[13]

3.3.3 When Optimality is not Relevant: Selecting "Representative Prompts". This strategy involves an appeal to the *ecological validity* of the study. When selecting prompts for a real-life setting, it is important to realize that *most people are not prompt experts*. So it is likely that the average user will provide suboptimal prompts (Zamfirescu-Pereira et al. 2023). If you are interested in the average-case performance of an LLM, you should also be using average prompts. These could be collected through a pilot study, where a representative set of potential users might interact with an LLM themselves, or where they are asked through a survey to write prompts that they might use for specified use cases.

12 One potential downside of continuous prompts is that we lose some of the interpretability that comes with human-readable prompts.
13 At some point we just have to fall back on an appeal to resource constraints.

3.4 The "Prior Research" Heuristic

There are two ways in which prior research can be used to establish the prompt for a new study: using exactly the same prompt as in earlier research, or phrasing a prompt based on examples or suggestions from the literature.

3.4.1 Exactly the Same Prompt. You could copy the exact prompt from an earlier study that performed the same task. This *might* yield good results, but it remains important to consider the context in which the prompt is used:

- Prompt performance might differ between models; a prompt that is optimal for one model may perform worse than its competitors when used with a different model (as, e.g., demonstrated in Schmidtová et al. 2025).
- Earlier results may have been carried out with a *different version* of the same model. It is not guaranteed that a for a given prompt the performance will remain stable across different model iterations.
- Generalizability of the findings of earlier studies depends on the reliability of those studies. (If a prompt was used in prior research, that does not automatically mean it is a good prompt.)

3.4.2 Conceptually the Same Prompt. Instead of using exactly the same prompt as in earlier research, one might also use a prompt that is *conceptually similar*. By this we mean that the prompt is either:

1. Based on best practices reported in earlier studies (e.g., Long et al. 2025), or:
2. Based on successful examples from earlier studies solving similar challenges.

The same limitations as in §3.4.1 apply here. Moreover, because there are likely to be alternative prompt formulations, one might still need to appeal to resource constraints/reasonable effort (§3.2).

3.5 No Justification

Finally, as with Lakens's (2022) list of justifications, one might honestly report not to have any justification for their choice of prompt. This might be seen as a weakness when you make strong claims about model performance, but different studies may have different goals. It does not always matter whether the authors are using "the perfect prompt" (if such a thing even exists).[14] We will now turn to some example goals that influence prompt design.

14 And as a side note: Ideally we would work towards models that are robust to prompt variation anyway.

4. Research Goals Shaping Prompt Design

Roughly speaking, the role that LLMs play in a study determines what properties (and what importance) their prompts ought to have, and therefore what kind of justification is required. A key distinction is whether the study is exploratory or confirmatory in nature (Schwab and Held 2020). Exploratory studies investigate a space of prompting choices in order to discover phenomena or generate hypotheses, and therefore benefit from prompt variation and transparency about design iterations. Confirmatory studies aim to make precise claims about model behavior or performance, and therefore require a stronger justification of the prompt as a controlled experimental condition. With this in mind, we identify four broad perspectives under which LLMs are commonly investigated in NLP and adjacent fields.[15]

1. **LLMs as Tools to Help Users Achieve a Task.** The focus in this paradigm is on whether an LLM can assist or automate a practical goal such as, e.g., summarization (Chuang et al. 2024), tutoring (Nye, Mee, and Core 2023), planning (Sun et al. 2024), or domain-specific retrieval (Zhuang et al. 2024). Prompts should approximate realistic interaction: Most users are not prompt engineers, and their inputs may be brief or suboptimal. Studies should therefore justify prompts in terms of plausibility, user-centered design, or performance under natural variation, potentially including representative prompt sampling or prompts sourced from prior practice.
2. **LLMs as Objects of Study.** When probing model capabilities (Liang et al. 2022, Section 8.2) or behavior (Chang and Bergen 2024), prompts act as experimental stimuli. Small prompt changes can produce large differences in output, making it difficult to separate model properties from prompt artifacts. Here, exploratory prompt variation is often necessary to discover how robust a finding is. Stronger prompt justification is required when making confirmatory claims about what a model can or cannot reliably do.
3. **LLMs as Measurement Instruments.** LLMs are increasingly used to rate or classify content (Li et al. 2024), as, e.g., judges of quality (Sedoc et al. 2025), factuality (Zhang and Gao 2023), or safety (Yuan et al. 2025). In such cases, prompts define the measurement protocol: what is being judged, according to which criteria, and in what format. Justification must therefore address validity (alignment with the intended construct), reliability (consistency across rephrasings), and transparency (reproducibility of the evaluation conditions).
4. **LLMs as Sociotechnical Actors.** In deployed applications—educational platforms (Chu et al. 2025), public services (Musumeci et al. 2024), creative tools (Dang et al. 2022), or decision-support pipelines (Eigner and Händler 2024)—prompt choices shape not only output quality but also users and institutions. Under this perspective, justification extends

15 These perspectives are not exhaustive nor mutually exclusive, but they capture the dominant motivations behind prompting choices in current research.

> to social consequences: safety, bias, robustness, and governance. Prompts generally encode normative assumptions about appropriate behavior,[16] and such assumptions should be made explicit.

Prompt choice is, therefore, both goal-dependent and tied to the evidentiary standard a study intends to meet, which in turn determines how rigorously prompts must be justified.

5. Future Research

It is too early to talk about "best practices" in prompt justification without the existence of an empirical evidence base. Although we believe that prompt development needs reporting standards, it is still unclear what those should look like, exactly. In the near term, it is useful for researchers to document their practices to enable future meta-scientific literature surveys to codify best practices. To this end, we offer some initial recommendations. If you do not do so already, a good first step is to keep a lab notebook to record your prompting history and the motivation behind each change. Best practices for keeping lab notebooks are discussed, for example, by Pain (2019) and Higgins, Nogiwa-Valdez, and Stevens (2022). When reporting your study, you should be able to answer the following questions:

- What kind of **justification** philosophy are you using and why?
- What **techniques** did you use to develop your prompt(s)?
- What **set of candidate prompts** did you select from?
- What **criteria** did you use to select your prompt(s)?
- How did candidate prompts **perform** with respect to those criteria?
- **What prompt(s) did you end up using?**

Capturing this information provides a minimum starting point, enabling readers to appreciate the generalizability of your findings and to understand how they relate to their own and others' research.

6. Final Note

At least for the foreseeable future, prompts are here to stay. The "naturalness" of this natural language interface is a double-edged sword: On the one hand, it made state-of-the-art technology accessible to the masses, but on the other hand, its brittleness poses serious risks to the generalizability of our results. Not just because small variations in the input can lead to serious performance degradation, but also because the intuitive interface makes it easy to forget the vast space of input parameters that we are manipulating through natural language. Thus, we need to make a conscious effort to make our prompting strategies explicit. As we have seen, earlier work has issued a call to provide justifications for all of our design choices; let's start by justifying our prompts!

16 These assumptions might either be explicit (stating how the LLM should behave) or implicit (omitting such instructions and relying on the default behavior of the model).

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